

Event-TimeRAF: AI Claim Verification Log

Verified against run 20260810T103436161252Z

Evidence contract. The executed verification notebook checks all 54 manifest entries by byte size and SHA-256, reconstructs metrics from 172,776 test forecast points, reruns the 2,000-resample paired comparisons and overlap-aware tests, and regenerates all nine manuscript figures. The immutable timestamped final-run ZIP is the numerical source of truth.

Claim	Checked how	Result
Run artifacts are unchanged and attributable	Recomputed byte size and SHA-256 for all 54 manifest entries	Confirmed
Final-publication profile was executed	Checked final flag, fresh Chronos gate, four stride-model runs, and completed manifest	Confirmed
48,332 windows; 34,020/7,113/7,199 chronological split	Reconstructed saved arrays and metadata; asserted shapes and counts	Confirmed
M04 reports MSE/MAE/RMSE/ R^2 of 26.185/3.125/5.117/0.379	Recomputed all metrics from 172,776 saved point predictions	Confirmed
Full M09 improves on M04	M09 MSE is 26.734 versus 26.185; paired interval is $[-0.122, 1.592]$	Corrected
Dense retrieval solves the full-model gap	Six-hour retrieval MSE is 34.616, but six-hour M09 is 26.412 versus M04 at 26.185	Corrected
The event channel improves retrieval	Weight 0.2 changes 91.4% of top- k sets but raises MSE from 38.080 to 38.411	Corrected
Chronos gain is retrieval-specific	M11–M10 is unresolved; climatology fusion 27.450 beats M11 28.733 by 1.283, CI [0.694, 1.913]	Corrected
Drift flag is an adaptive gate	Code and artifacts show a diagnostic flag only; it does not select or update a model	Corrected
Storm-event context is available in strict real time	NOAA publication timestamps are absent; event results are retrospective sensitivity analyses	Flagged unverified
Findings generalize geographically and runtime is hardware-normalized	Only Los Angeles County is evaluated and hardware identifiers are absent	Flagged unverified

AI writing removed or corrected (before → after).

Retrieval claim	“Retrieval gives selective gains”	Dense retrieval helps retrieval-only, but not the best feature model; M11 is worse than climatology fusion
Event claim	Event context presented as a forecasting contribution	Event ranking is active but degrades the tested retrieval forecast
Drift claim	Drift-aware wording implied adaptation	Upper-tail score retained only as a diagnostic and evidence field
Claim scope	Event-aware findings phrased as operational evidence	Retrospective timing, one-county validation, and non-normalized runtime stated explicitly